Part A – MCQs

1. Traditional Programming follows:
   (**b) Rules**
2. Machine Learning learns from:
   (**b) Data**
3. Transformer introduced:
   (**c) Attention**
4. GPT mainly performs:
   (**b) Next-token prediction**
5. Llama is:
   (**b) Open Model**

Part B – Short Answers

1. Explain Artificial Intelligence with examples.

Artificial Intelligence (AI) is the ability of computers to perform tasks that normally require human intelligence, such as understanding language, recognising images, and making decisions. Examples include ChatGPT, Google Maps, Siri, Alexa, and self-driving cars.

2. Differentiate Traditional Programming and Machine Learning.

- Traditional Programming: Uses fixed rules written by a programmer.
- Machine Learning: Learns patterns from data and improves predictions without explicit rules.

3. What is Deep Learning?

Deep Learning is a type of Machine Learning that uses neural networks with many layers to solve complex tasks like image recognition, speech recognition, and language translation.

4. Explain Transformers.

Transformers are AI models that use an attention mechanism to understand relationships between words. They are the technology behind models like GPT and Llama.

5. What are Foundation Models?

Foundation Models are large AI models trained on huge amounts of data. They can perform many tasks such as writing, translating, summarising, and answering questions.

6. Open vs Closed Models.

- **Open Models:** Publicly available, can often be modified and used by developers (e.g., Llama).
- **Closed Models:** Proprietary models controlled by a company and not fully open (e.g., GPT).

Part C – Practical Activities

Activity 1: Compare responses from two AI tools

Prompt: Explain AI to a 12-year-old.

ChatGPT Response

AI is like a smart computer helper that learns from data and helps people answer questions, write stories, solve maths, and recognise pictures.

Gemini Response

AI is a technology that helps computers think and learn from examples. It can be used in voice assistants, games, and education.

Comparison

- **Response Quality:** ChatGPT is more detailed.
- **Clarity:** Both are easy to understand.
- **Examples Used:** ChatGPT gives better examples.
- **Response Length:** ChatGPT is longer; Gemini is shorter.

Conclusion

Both ChatGPT and Gemini explain AI clearly. ChatGPT provides more detailed explanations with useful examples, making it easier for beginners to understand. Gemini gives shorter and simpler answers, which are quick to read. ChatGPT is better for learning concepts in depth, while Gemini is helpful for getting quick information. Both are useful AI tools, but ChatGPT offers more detailed educational content.

Activity 2 – Prompt Engineering

Prompt 1

Explain Machine Learning.

Prompt 2

Explain Machine Learning in simple language.

Prompt 3

Explain Machine Learning using a real-life example.

Prompt 4

Explain Machine Learning as if I am a Class 8 student.

Five Observations

1. Better prompts give better answers.
2. Adding audience details changes the response.
3. Examples make explanations easier.
4. Simple language improves understanding.
5. Specific prompts produce more accurate results.

Activity 3 – 10 AI Applications

Application	AI Feature	User Benefit
ChatGPT	Text generation	Answers questions
Gemini	AI assistant	Learning and writing
Google Maps	Route prediction	Faster navigation
YouTube	Recommendations	Better video suggestions
Netflix	Recommendations	Personalised content

Application	AI Feature	User Benefit
Spotify	Music recommendation	Better playlists
Grammarly	Grammar correction	Better writing
Canva	AI design	Easy graphics creation
Google Translate	Language translation	Easy communication
Alexa	Voice assistant	Hands-free help

Part D – Research Assignment

1. Compare GPT, Gemini, Claude and Llama

Model	Developer	Open/Closed	Strength	Use Case
GPT	OpenAI	Closed	Writing & coding	Chatbots
Gemini	Google	Closed	Google integration	Search & productivity
Claude	Anthropic	Closed	Long documents	Research
Llama	Meta	Open	Customisation	AI development

2. Three applications I use regularly

1. **ChatGPT** – Helps me learn new topics and write content.
2. **Google Maps** – Finds the best routes and traffic updates.
3. **YouTube** – Recommends useful educational videos.

Part E – Reflection

How will Generative AI change software development and everyday life over the next five years?

Generative AI will have a major impact on software development and daily life. Developers will use AI to write code, find bugs, create documentation, and improve productivity. This will reduce development time and help beginners learn programming more easily.

In everyday life, AI will help people write emails, translate languages, create images, summarise documents, and answer questions quickly. Students will use AI for learning, teachers will prepare lessons faster, and businesses will improve customer support using AI assistants.

Healthcare may benefit from AI by helping doctors analyse medical data. In education, AI can provide personalised learning experiences. However, AI should be used responsibly because it can sometimes produce incorrect information. Human review and ethical use are important.

Overall, Generative AI will make work faster, improve creativity, and simplify many daily tasks. It is expected to become an essential tool for education, business, and software development in the coming years.